EXP 2	**Develop a program to perform Bias Audit on pre-trained word embeddings to identify gender or racial stereotypes using the Word Embedding Association Test (WEAT).**

AIM:

To develop a program that performs a **Bias Audit on pre-trained word embeddings** using the **Word Embedding Association Test (WEAT)** to identify **gender stereotypes** present in **GloVe word embeddings**. The system measures the differential association between male/female words and career/family words using cosine similarity, WEAT score, and effect size to quantify the degree of bias.

PROCEDURE:

1. **Install Dependencies:** Install the gensim library using pip and import required libraries such as numpy and gensim.downloader.
2. **Load Pre-trained Embeddings:** Load the GloVe word embeddings (glove-wiki-gigaword-100) using the gensim downloader API.
3. **Define Word Sets:** Define four word sets — Male words, Female words, Career words, and Family words — to be used in the WEAT test.
4. **Filter Valid Words:** Filter each word set to retain only words present in the loaded embedding model vocabulary.
5. **Compute Cosine Similarity:** Define a cosine similarity function to measure the closeness between two word vectors using dot product and norms.
6. **Compute Association Score:** For each target word, calculate the mean cosine similarity with attribute set A and attribute set B, then find the differential association.
7. **Compute WEAT Score:** Calculate the WEAT score as the difference between the sum of associations of target set X and target set Y with attribute sets A and B.
8. **Compute Effect Size:** Calculate the effect size by dividing the difference in mean associations of X and Y by the standard deviation of the combined association scores.
9. **Analyze and Report Bias:** Print the WEAT score and effect size, and interpret the direction of bias — whether male words associate more with career or female words associate more with family.

CODE:

```
!pip install gensim
import numpy as np
import gensim.downloader as api

print("Loading GloVe embeddings...")
model = api.load("glove-wiki-gigaword-100")
print("Embeddings loaded!")

male_words   = ["man","male","boy","brother","he","him","his","son"]
female_words = ["woman","female","girl","sister","she","her","hers","daughter"]

career_words = ["executive","management","professional",
                "corporation","salary","office","business","career"]

family_words = ["home","parents","children",
                "family","cousins","marriage","wedding","relatives"]

def cosine_similarity(a, b):
    return np.dot(a, b) / (np.linalg.norm(a) * np.linalg.norm(b))

def association(word, A, B):
    w    = model[word]
    simA = np.mean([cosine_similarity(w, model[a]) for a in A])
    simB = np.mean([cosine_similarity(w, model[b]) for b in B])
```

```
    return simA - simB

def weat_score(X, Y, A, B):
    scoreX = sum(association(x, A, B) for x in X)
    scoreY = sum(association(y, A, B) for y in Y)
    return scoreX - scoreY

def effect_size(X, Y, A, B):
    assocX    = np.array([association(x, A, B) for x in X])
    assocY    = np.array([association(y, A, B) for y in Y])
    numerator = np.mean(assocX) - np.mean(assocY)
    combined  = np.concatenate([assocX, assocY])
    return numerator / np.std(combined)

def filter_words(words):
    return [w for w in words if w in model]

male_words   = filter_words(male_words)
female_words = filter_words(female_words)
career_words = filter_words(career_words)
family_words = filter_words(family_words)

print("\nRunning WEAT Bias Test...\n")

score  = weat_score(male_words, female_words, career_words, family_words)
effect = effect_size(male_words, female_words, career_words, family_words)

print("Male Words:  ", male_words)
print("Female Words:", female_words)
print("\nCareer Words:", career_words)
print("Family Words:", family_words)

print("\nWEAT Score:", score)
print("Effect Size:", effect)

if effect > 0:
    print("\nBias Detected:")
    print("Male words are more associated with career")
    print("Female words are more associated with family")
else:
    print("\nBias Detected:")
    print("Female words are more associated with career")
    print("Male words are more associated with family")
```

Output:

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.whl (27.9 MB)
   ---------------------------------------- 27.9/27.9 MB 78.9 MB/s eta 0:00:00
Successfully installed gensim-4.4.0
Loading GloVe embeddings...
[==================================================] 100.0% 128.1/128.1MB downloaded
Embeddings loaded!

Running WEAT Bias Test...

Male Words: ['man', 'male', 'boy', 'brother', 'he', 'him', 'his', 'son']
Female Words: ['woman', 'female', 'girl', 'sister', 'she', 'her', 'hers', 'daughter']

Career Words: ['executive', 'management', 'professional', 'corporation',
               'salary', 'office', 'business', 'career']
Family Words: ['home', 'parents', 'children', 'family',
               'cousins', 'marriage', 'wedding', 'relatives']

WEAT Score:  0.81999767
Effect Size: 0.95399064
```

```
Bias Detected:
Male words are more associated with career
Female words are more associated with family
```

Result:

The Bias Audit program was successfully implemented using **GloVe pre-trained word embeddings** and the **Word Embedding Association Test (WEAT)**. The test measured differential associations between gender word sets and attribute word sets using cosine similarity.

The computed **WEAT Score of 0.81999767** and **Effect Size of 0.95399064** confirm the presence of gender bias in the embeddings. The results show that **male words are more associated with career-related terms** while **female words are more associated with family-related terms**, demonstrating how pre-trained word embeddings can capture and reinforce existing societal stereotypes.